Quantum Fault Tolerance Meets Toom's Contours

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May 17, 2026

1 Shrinking

We are going to show that the SWEEP decoder 'shrinks' the error by proving that the edges/poles of error clusters get closer as a result of a decoding step. For that, we consider the standard embedding of the grid in $\mathbb{R}^d$ and use the term of potential walls.

Definition 1.1 (The standart embedding)**.** Consider the d-dimensional infinite grid $\mathcal{G}^d$. The *standard embedding* $\mathcal{G}^d \to \mathbb{R}^d$ is given by fixing a vertex in $\mathcal{G}^d$ to be sent to the origin and then mapping the generators of the grid to the standard basis. We extend notation as follows: for a vertex $v \in \mathcal{G}^d$, a generator $g \in S$, and a coordinate $i \in [d]$, such that under the embedding $g \to i$, we denote by $v_i = v_g$ the i'th coordinate of the image of v.

Definition 1.2 (Potential Walls)**.** We define the *potential walls* to be the $d+1$ functions $L_1, L_2, .., L_{d+1}$, from the vertices of $\mathcal{G}_n^d$ to the reals as follow:

1. For any coordinate $i \in [d]$, we define $L_i \colon \mathcal{G}^d \to \mathbb{R}$ as $L_i(v) := v_i$. In addition, for the generator $g \in S$ which is mapped to i, i.e $g \to i$, we identify $L_g = L_i$.
2. We define $L_{d+1} \colon \mathcal{G}^d \to \mathbb{R}$ by $L_{d+1}(v) := \sum_{i=1}^{d} v_i$.

Consider an assignment z over the faces. For a vertex $v \in z$, and a potential wall $l \in \{L_1, .., L_{d+1}\}$, we say that v is a *local maximum* of l in z if for any neighbor u of v, such that $u \in z$, we have $l(v) \geq l(u)$. We say that v is a *strong local maximum* if the inequality is strict, namely $l(v) > l(u)$. When the assignment z is clear from the context, we might omit it and briefly say that v is a local maximum of l.

With the assistance of potential walls, we can discuss the endpoint of a bunch of bits, which will soon become error clusters. The following two claims relate the local maximum points of $L_1, \ldots, L_d$ in an assignment z to the structure of the syndrome they observe. Later, we will use that relation to infer that the decoding step of the SWEEP rule can't spread the error to a vertex[1] with a higher value than the current global maximum of $L_1, \ldots, L_d$. In other words, the error does not spread in the positive directions.

[1] To faces whose vertices have higher values of $L_1, \ldots, L_d$ than the current global maximum values.

Claim 1.3. Let z be an assignment, and recall that the vertices in ∂z are the set of vertices adjacent to unsatisfied checks induced by z. Let v be a vertex in ∂z, and let g be a generator in S such that v is a local maximum of L_g in ∂z. Then $gv \notin \partial z$.

Proof. Suppose $gv \in \partial z$, then

$$L_g(gv) = (gv)_g = 1 + v_g = 1 + L_g(v)$$

In contradiction to v being a local maximum of L_g in ∂z. $\square$

Claim 1.4. Let z be an assignment, let v be a vertex in ∂z, and let g be a generator in S such that v is a local maximum of L_g. Then, for any check associated with a $d' - 1$ face rooted at v that contains gv, it is satisfied.

Proof. Assume, through contradiction, the existence of an unsatisfied check associated with a $d' - 1$ positive face rooted at v, which contains gv. This implies $gv \in \partial z$, contradicting Claim 1.3. $\square$

For the $(d+1)$th potential wall, we would like to have a stronger claim that implies the error cluster shrinks. For this, we start by showing that any error is equivalent, up to stabilizer addition, to an error whose left-corner edge sees a syndrome. We present two notations of local codes that describe the local view of a $(d+1)$th edge that sees no syndrome. They are the *stabilizers code at vertex* v and the *small code at vertex* v. The first corresponds to stabilizers rooted at the vertex, while the second is the collection of local views on its positive faces that satisfy its positive checks. Then, in Claim 1.7, we consider a vertex which is a $(d+1)$th edge and doesn't see a syndrome, meaning that its local view is in its *small code*, and show that there is a codeword of its *stabilizers code* that agrees with its positive faces. Addition by that stabilizer yields a new assignment excluding v.

Definition 1.5 (Stabilizers Code at Vertex v)**.** We define *stabilizer code at vertex* v, denoted by Stab_v, as the restriction of the (d, d')-Toric code to the d'-dimensional faces such that their vertices are at a distance of at most one positive step in each direction from v. That is equivalent to the union of all the positive d'-dimensional faces of v with the d-dimensional faces of each of its sibling gv, excluding faces containing g as their generator. Namely:

$$\begin{aligned}\mathrm{Stab}_v := \big\{z : \partial^- z = 0 \text{ and } \\ z \in \mathrm{Span}\big(\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d'\} \\ \bigcup_{g \in S} \{\theta_{\mathbf{gv},\mathbf{S'/g}} : S' \subset S/g, |S'| = d'\}\big)\big\}\end{aligned}$$

The checks of the code, are of the following three types:

1. Checks that are rooted at v, each identified by choosing $d - 1$ generators from S. Namely:

$$\{\theta_{\mathbf{v},\mathbf{S'}} : S' \subset S, |S'| = d' - 1\}$$

For each check, the bits that connect to it, are given by restricting the coboundary of the check to bits domain of Stab_v:

$$\big(\partial\theta_{\mathbf{v},\mathbf{S'}}\big)|_{\mathrm{Stab}_v} = \big(\sum_{g\in S/S'}\theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}} + \sum_{g\in S'}\theta_{\mathbf{g^{-1}v},\mathbf{S'}\cup\mathbf{g}}\big)|_{\mathrm{Stab}_v} = \sum_{g\in S/S'}\theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}}$$

2. Checks are rooted at the positive sibling of v. Consider such a sibling, let's say gv (namely the vertex that is given by stepping in the direction $g\in S$ from v). Notice that any face rooted at gv that contains ggv cannot be contained in faces rooted at v. Thus, the checks of gv correspond to choosing subsets from S/g. Namely:

$$\bigcup_{g\in S}\big\{\theta_{\mathbf{gv},\mathbf{S'}} : S'\subset S/g, |S'| = d'-1\big\}$$

And each check is supported by:

$$\begin{aligned}\big(\partial\theta_{\mathbf{gv},\mathbf{S'}}\big)|_{\mathrm{Stab}_v} &= \big(\sum_{g'\in S/S'}\theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g'}} + \sum_{g'\in S'}\theta_{\mathbf{g'^{-1}v},\mathbf{S'}\cup\mathbf{g'}}\big)|_{\mathrm{Stab}_v} \\ &= \sum_{g'\in S/S'}\theta_{\mathbf{gv},\mathbf{S'}\cup\mathbf{g}} + \theta_{\mathbf{g^{-1}v},\mathbf{S'}\cup\mathbf{g}}\end{aligned}$$

3. Checks that their roots are at a two-step distance from v, namely vertices of the form $g,g'\in S$, where $g\neq g'$. By the arguments presented in the second type, the checks are the $d'-1$ faces which do not have the generators on the path leading from v to them, i.e., g,g', in their generator set.

$$\bigcup_{g,g' inS'}\big\{\theta_{\mathbf{g'gv},\mathbf{S'}} : S'\subset S/\{g,g'\}, |S'| = d'-1\big\}$$

And for each check, its support is:

$$\begin{aligned}\big(\partial\theta_{\mathbf{g'gv},\mathbf{S'}}\big)|_{\mathrm{Stab}_v} &= \big(\sum_{g\in S/S'}\theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}} + \sum_{g\in S'}\theta_{\mathbf{g^{-1}v},\mathbf{S'}\cup\mathbf{g}}\big)|_{\mathrm{Stab}_v} \\ &= \sum_{g\in S/S'}\theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}} + \sum_{g\in S'}\theta_{\mathbf{g^{-1}v},\mathbf{S'}\cup\mathbf{g}}\end{aligned}$$

Definition 1.6 (Small Code at Vertex v)**.** For a vertex v, we denote by $\mathcal{C}_v$ the *small code at vertex* v, which is defined to be the binary assignments over the bits (d' faces) rooted at v that satisfy the positive checks of v, namely:

$$\begin{aligned}\mathcal{C}_v := \big\{z : \partial^- z = 0 \text{ and} \\ z\in \mathrm{Span}\big(\{\theta_{\mathbf{v},\mathbf{S'}} : S'\subset S, |S'| = d'\}\big)\big\}\end{aligned}$$

The checks of the code are exactly the first type checks of Stab_v.

Claim 1.7. Let v be a vertex. For any $z \in \mathcal{C}_v$, there is a stabilizer $w \in \text{Stab}_v$ such that $z = w|_v$; namely, z and w agree on the faces rooted at v.

Proof. We argue that the restriction of the generating matrix of Stab_v equals the generating matrix of $\mathcal{C}_v$. This means that any codeword of Stab_v is an extension of a codeword of $\mathcal{C}_v$.

□

Claim 1.8. Let z be a finite assignment. Suppose that

$$\max\left\{L_{d+1}(u) : u \in z\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1$$

Proof. Let $v_1, v_2, \ldots, v_k$ be the vertices in z that achieve the maximum for L_{d+1}, namely $v_1, v_2, \ldots, v_k = \arg\max_{v \in z} L_{d+1}(v)$, and assume that $v_i \notin \partial^{\pm} z$ for any $i \in [k]$.

Consider the ith vertex v_i. By definition, since $v_i \notin \partial^{\pm} z$, we have $\partial^{\pm} z|_{v_i} = 0$, and since v_i is a maximum point of L_{d+1} in z, it is also a local maximum, and therefore $z|_{v_i +} = z|_{v_i}$. Combining the two, we get that $z|_{v_i}$ is a codeword of the small code at v_i, namely $z|_{v_i} \in \mathcal{C}^{\pm}_{v_i}$. Thus, by Claim 1.7, there is a stabilizer $w_i \in \text{Stab}^{\pm}_{v_i}$ such that w_i and z agree on the positive faces of v_i, namely $w_i|_{v_i} = z|_{v_i}$.

Set $\tilde{z} \leftarrow z + \sum_i w_i$. Since $\sum_i w_i$ is a stabilizer, we have $\tilde{z} \sim_{\mathcal{C}_X} z$. In addition, for any $j \neq i$, it follows that $v_j \notin w_i$. Otherwise, there exists $S' \subset S$ such that $v_j \in \theta_{v_i, S'}$, and therefore $L_{d+1}(v_j) > L_{d+1}(v_i)$, in contradiction to v_i and v_j being both maximum points for L_{d+1} in $\partial^{\pm} z$. Thus, we have:

$$\tilde{z}|_{v_i} = (z + \sum_i w_i)|_{v_i} + (z + w_i)|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{z}$. Hence, the vertex domain of $\tilde{z}$ is contained in the union of the vertices of z with the vertices of $w_1, w_2, \ldots, w_k$, substituting $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} &\leq \max\left\{L_{d+1}(u) : u \in z \bigcup_i w_i / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_{d+1}(u) : u \in z\right\} - 1\end{aligned}$$

□

Claim 1.9. Let z be a finite assignment. Then there is $\tilde{z} \sim_{\mathcal{C}_X} z$ such that

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\}$$

Proof. Let z be a finite assignment, and let k be an integer such that $k \geq 1$ and

$$\max\left\{L_{d+1}(u) : u \in z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm} z\right\} + k$$

For an integer $l \leq k$, consider the assignments $\tilde{z}_0, \tilde{z}_1, .., \tilde{z}_l$ defined as follow:

- The first element, $\tilde{z}_0 = z$.
- If the $i-1$th element satisfies

$$(\star) \quad \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} > \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}_{i-1}\right\}$$

  Then define the ith element, $\tilde{z}_i$, to be the result of applying the construction, given in Claim 1.8, on $\tilde{z}_{i-1}$, namely $\partial\tilde{z}_{i+1} = \partial z_i$ and

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}_i\right\} \leq \max\left\{L_{d+1}(u) : u \in \tilde{z}_{i-1}\right\} - 1$$

  If the $i-1$th element doesn't satisfies $\star$, then set $l = i-1$.
- If the kth element was defined, set $l = k$.

Since $\sim_{C_X}$ is a transitive relation, we have that for any $i \in [l]$, $z \sim_{C_X} \tilde{z}_i$. If $l \neq k$, then there exists $\tilde{z}_i$ such that $\tilde{z}_i \sim_{C_X}$ and $\tilde{z}_i$ satisfies $\star$, namely by setting $\tilde{z} = \tilde{z}_i$, we get the desired. So, it is left to handle the case where $l = k$. In that case, set $\tilde{z} = \tilde{z}_k$. By Claim 1.8:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in \partial^{\pm}z\right\}$$

Yet, since $\tilde{z} \sim_{C_X} z$, it follows that $\partial\tilde{z} = \partial z$, So:

$$\max\left\{L_{d+1}(u) : u \in \partial^{\pm}z\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}\right\} \leq \max\left\{L_{d+1}(u) : u \in z\right\}$$

Therefore, we have the equality:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}z\right\}$$

□

Claim 1.10. Let z be a finite assignment and let ξ be the result of the application of the Sweep rule on it. Then:

1. For any $i \in [d]$: $\max\left\{L_i(v) : v \in \partial z\right\} = \max\left\{L_i(v) : v \in \partial\xi\right\}$
2. And for $i = d+1$: $\max\left\{L_{d+1}(v) : v \in \partial z\right\} > \max\left\{L_{d+1}(v) : v \in \partial\xi\right\} + 1$

Proof. First, since the Sweep rule is invariant to addition by stabilizers, for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, the correction applied by the Sweep rule on $\tilde{z}$ is the same as the application on z. Denote by ϕ the correction, by $\tilde{\xi}$ the result of applying the Sweep rule on $\tilde{z}$, and by w the stabilizer which is the difference between z and $\tilde{z}$, namely:

$$\xi = z + \phi$$
$$\tilde{\xi} = \tilde{z} + \phi$$
$$\tilde{z} = z + w$$

The $\partial^{\pm}$ boundary of $\tilde{\xi}$ equals:

$$\partial\tilde{\xi} = \partial\left(\tilde{z}+\phi\right) = \partial\left(z+\phi\right) = \partial\xi$$

So, in total, we have that for any $\tilde{z}$ such that $\tilde{z} \sim_{C_X} z$, it holds that $\partial\tilde{z} = \partial z$ and $\partial\tilde{\xi} = \partial\xi$. Therefore, showing that there exists $\tilde{z} \sim_{C_X} z$ for which the claim holds proves the claim for z. By Claim 1.9, there is $\tilde{z} \sim_{C_X} z$ such that:

$$\max\left\{L_{d+1}(u) : u \in \tilde{z}\right\} = \max\left\{L_{d+1}(u) : u \in \partial^{\pm}\tilde{z}\right\}$$

For such $\tilde{z}$, any maximum of L_{d+1} in $\partial\tilde{z}$, is also a maximum in $\tilde{z}$. Denote by $v_1, .., v_k$ the maximum points in $\partial\tilde{z}$. Since they are maximum points in $\tilde{z}$, it holds that for each $i \in [k]$, $\tilde{z}|_{v_i} = \tilde{z}|_{v_i+}$, and hence $\partial\tilde{z}|_{v_i} = \partial\tilde{z}|_{v_i+}$. Since $\tilde{z}|_{v_i+}$ is an assignment rooted at v_i, then v_i can suggest $\tilde{z}|_{v_i+}$, namely the condtion in Sweep rule is satisfied for vertex v_i. Thus:

$$\partial\tilde{\xi}|_{v_i} = \partial\tilde{z}|_{v_i} + \partial\phi|_{v_i} = 0$$

Namely, for any i, the ith vertex v_i is not in $\tilde{\xi}$. For any vertex v denote by ϕ_v the correction made by the vertex v, namely $\phi = \sum_{v\in\partial\tilde{z}}\phi_v$. Hence, the vertex domain of $\tilde{\xi}$ is contained in the union of the vertices of $\tilde{z}$ with the vertices of $\cup_{v\in V}\phi_v$, substituting maximum points $v_1, v_2, \ldots, v_k$. Overall, we get:

$$\begin{aligned}\max\left\{L_{d+1}(u) : u \in \partial\tilde{\xi}\right\} &\leq \max\left\{L_{d+1}(u) : u \in \partial\tilde{z} \bigcup_{v\in\partial\tilde{z}} \phi_v / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_{d+1}(u) : u \in \partial\tilde{z}\right\} - 1\end{aligned}$$

Let v be a vertex that is a maximum of L_g in $\partial\tilde{z}$, maximum points of L_g do not see syndrome at the g direction. If and:

$$\phi_v = \sum \theta_{\mathbf{v},\mathbf{S'_i}}$$

By Claim 1.4 $\partial\tilde{z}|_{gv} = 0$, Hence:

$$\begin{aligned}\partial\phi_v|_{gv} &= \left(\partial\sum\theta_{\mathbf{v},\mathbf{S'_i}}\right)|_{gv} \\ &= \left(\sum_{S'_i}\sum_{g'\in S'_i}\theta_{\mathbf{v},\mathbf{S'_i/g'}} + \theta_{\mathbf{g'v},\mathbf{S'_i/g'}}\right)|_{gv} \\ &= 0\end{aligned}$$

Thus if $g \in \cup S'_i$, Then:

$$\theta_{\mathbf{gv},\mathbf{S'_i/g}} \in \partial\phi_v|_{gv}$$

So for $\theta_{\mathbf{gv},\mathbf{S'_i/g}} = 0$, it must appear at least twice in the sum, yet the left terms in the sum cannot contain it (they differ by the root vertex), and if there is a right term

$\theta_{\mathbf{gv},\mathbf{S'_j/g}} = \theta_{\mathbf{gv},\mathbf{S'_i/g}}$, then it implies that $S_j = S_i$. Which is a contradiction, therefore: $g \notin \cup S'_i$ and $gv \notin \phi_v$.

$$\begin{aligned}\max\left\{L_g(u) : u \in \partial\tilde{\xi}\right\} &\leq \max\left\{L_g(u) : u \in \partial\tilde{z} \bigcup_{v \in \partial\tilde{z}} \phi_v / \bigcup_i v_i\right\} \\ &\leq \max\left\{L_g(u) : u \in \partial\tilde{z}\right\} + 1 - 1\end{aligned}$$

□

1.1 Drafts-Garbage

[COMMENT] Regarding the Claim 1.7, The number of restriction in ∂^+ case was counted in negligently. In addition, we should show that for the ∂^- case, restrictions of the form $\theta_{\mathbf{g_2g_1v},\mathbf{S}}$ are degenerate. In the bottom line it's look like:

$$\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S}} = \partial^+\theta_{\mathbf{g_1v},(\mathbf{S}/\mathbf{g_3})\cup\mathbf{g_2}} + \partial^+\theta_{\mathbf{g_2v},(\mathbf{S}/\mathbf{g_3})\cup\mathbf{g_1}}$$

Lets develop it, consider a restriction rooted at g_2g_1v, and let's look on it's boundary:

$$\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S'}} = \sum_{g \in S/S'} \theta_{\mathbf{g_2g_1v},\mathbf{S'}\cup\mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g^{-1}g_2g_1v},\mathbf{S'}\cup\mathbf{g}}$$

Now, any term in the left closure is a bit beyond the scope of the bits in Stab_v^-, thus, we are zeroing them out. The bits, on the right closers are in the scope, only if $g \in \{g_1, g_2$. Thus, in total the check is equivalence to:

$$\begin{aligned}\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S'}} &= \theta_{\mathbf{g_1v},\mathbf{S'}\cup\mathbf{g_2}} \cdot \mathbf{1}_{g_2 \notin S'} \\ &+ \theta_{\mathbf{g_2v},\mathbf{S'}\cup\mathbf{g_1}} \cdot \mathbf{1}_{g_1 \notin S'}\end{aligned}$$

$$\partial^+\theta_{\mathbf{g_1v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_2}} = \sum_{g \in S/S'} \theta_{\mathbf{g_1v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_2}\cup\mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g^{-1}g_1v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_2}\cup\mathbf{g}}$$

$$\partial^+\theta_{\mathbf{g_2v},(\mathbf{S}/\mathbf{g_3})\cup\mathbf{g_1}} = \sum_{g \in S/S'} \theta_{\mathbf{g_2v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_1}\cup\mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{g^{-1}g_2v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_1}\cup\mathbf{g}}$$

$$\partial^+\theta_{\mathbf{g_1v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_2}} = \left(\sum_{g \in S/S'} \theta_{\mathbf{g_1v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_2}\cup\mathbf{g}}\right) + \theta_{\mathbf{v},(\mathbf{S'}/\mathbf{g_3})\cup\{\mathbf{g_2},\mathbf{g_1}\}}$$

$$\partial^+\theta_{\mathbf{g_2v},(\mathbf{S}/\mathbf{g_3})\cup\mathbf{g_1}} = \left(\sum_{g \in S/S'} \theta_{\mathbf{g_2v},(\mathbf{S'}/\mathbf{g_3})\cup\mathbf{g_1}\cup\mathbf{g}}\right) + \theta_{\mathbf{v},(\mathbf{S'}/\mathbf{g_3})\cup\{\mathbf{g_1},\mathbf{g_2}\}}$$

Other way:

$$\begin{aligned}\partial^+ \sum_{g' \in S/S'} \theta_{\mathbf{g'v},\mathbf{S'}} &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g'v},\mathbf{S'}\cup\mathbf{g}} + \sum_{g \in S/S'} \theta_{\mathbf{v},\mathbf{S'}\cup\mathbf{g}} \\ &= \sum_{g' \neq g \in S/S'} \theta_{\mathbf{g'v},\mathbf{S}\cup\mathbf{g}} + \partial^+\theta_{\mathbf{v},\mathbf{S'}}\end{aligned}$$

Where we use the fact that if $g = g'$ then the bit is beyond the scope. In addition:

$$\begin{aligned}\partial^+ \sum_{g'<g\in S/S'} \theta_{\mathbf{g'gv},\mathbf{S'}} &= \sum_{g'<g\in S/S'} \theta_{\mathbf{g'v},\mathbf{S'\cup g}} + \theta_{\mathbf{gv},\mathbf{S'\cup g'}} \\ &= \sum_{g'\neq g\in S/S'} \theta_{\mathbf{g'v},\mathbf{S'\cup g}} \\ &= \partial^+ \sum_{g'\in S/S'} \theta_{\mathbf{g'v},\mathbf{S'}} + \partial^+\theta_{\mathbf{v},\mathbf{S'}}\end{aligned}$$

Thus, we find more $\binom{d}{d'-1}$ restriction over the checks. [We need to show that those restrictions are independent]. In addition:

$$\partial^+ \sum_{S'} \sum_{g'<g\in S/S'} \theta_{\mathbf{g'gv},\mathbf{S'}} = 0$$

$$\partial^+\theta_{\mathbf{g_2g_1v},\mathbf{S}} = \partial^+\theta_{\mathbf{g_1v},(\mathbf{S}/\mathbf{g_3})\cup\mathbf{g_2}} + \partial^+\theta_{\mathbf{g_2v},(\mathbf{S}/\mathbf{g_3})\cup\mathbf{g_1}}$$

I don't succes to advance the idea more than that, So in the end I believe that I will do is to say that the diminsion of Stab_v^- is at least $\binom{d}{d'+1}$. And by fixing coordinates we add more $\binom{d}{d'}$ restriction. So the dimension becomes:

$$\binom{d}{d'+1} - \binom{d}{d'}$$

$$\binom{d}{d'+1} - \binom{d}{d'} + \binom{d}{d'-1}$$

Note that since we talk on mapping checks $\to$ bits, we use the ∂^+. Eventually, we need to use the poincare duality $H^{d'} \to H_{d-d'}$, the duality also should be used to define the SWEEP rule to C_Z. I didn't successes to do it other way.

A wrong proof of Claim 1.7

Proof. We will show that codes obtained from Stab_v by forcing the faces rooted at v to $z|_v$ have non-zero dimensions. Then it follows that any code word of $\mathcal{C}_v$ could be extend to codewords in Stab_v. The number of bits in both Stab_v is:

$$\binom{d}{d'} + d\binom{d-1}{d'}$$

The numbers of checks for $\mathrm{Stab}_\pm$ are:

$$\binom{d}{d'\pm 1} + d\binom{d-1}{d'\pm 1}$$

The restriction to $z|_v$ sets the bits rooted at v and satisfies the checks that are rooted at v, so the checks of the form $\theta_{\mathbf{v},\mathbf{S'}}$ for $|S'| = d' \pm 1$ are degenerate, and the dimension that remains is:

$$\dim \geq d\binom{d-1}{d'} - d\binom{d-1}{d'\pm 1}$$

In the case that $d' = (d-1)/2$, then for any $x \in [d-1]$ we have $\binom{d-1}{d'} > \binom{d-1}{x}$ therefore:

$$\begin{aligned}\dim &\geq d\binom{d-1}{d/2} - d\binom{d-1}{d/2 \pm 1} \\ &> d\end{aligned}$$

So there is at least 2^d stabilizers $w \in \mathrm{Stab}_v$ that agree with z. $\square$